04 · Which funnel stage to invest in? — mediation (pathmc)

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The business decision. We have a fixed budget to split across the growth funnel — **onboarding -> engagement -> activation -> conversion** — and want to put it where it moves paid conversion the most. The complication is that the stages form a **chain**: money spent improving onboarding doesn't only help onboarding, it *cascades* downstream into engagement, activation, and ultimately conversion. So we can't just correlate each stage with revenue; we have to trace the causal *paths* and decide where a push travels furthest.

1.0.1 The concept: mediation, direct and indirect effects

When a treatment X affects an outcome Y *through* an intermediate variable M (a **mediator**), the total effect splits in two:

- the **direct effect** — X's influence on Y that does *not* go through M (Pearl's *natural direct effect*, NDE);
- the **indirect effect** — the part that flows $X \to M \to Y$ (the *natural indirect effect*, NIE).

Total effect = direct + indirect, and the **proportion mediated** = indirect / total tells you how much of a stage's value is really realised *downstream*. In a linear model the indirect effect is just the **product of the coefficients along the path** (the classic "a · b" of mediation analysis) — `pathmc` generalises this by pushing posterior draws through the intervened graph, so we get every path's effect *with uncertainty*.

1.0.2 The catch this notebook is honest about

Mediation leans on a **stronger** assumption than a simple treatment-effect analysis: no unmeasured confounding of the *mediator–outcome* link (on top of the usual treatment–outcome one). If some unmeasured trait drives both engagement *and* conversion, the direct/indirect *split* is biased even when the *total* effect is fine. So Step 6 doesn't just report the split — it **stress-tests how much such confounding would have to exist to flip the "invest here" decision.**

> **On real data.** Swap in your **product-analytics event data** — per-user onboarding score, engagement, activation flag, conversion — which every growth team already logs. The method needs the funnel's causal *order* to be known (it usually is) and the no-hidden-mediator-confounder assumption to be defensible (Step 6 shows how to argue it).

`pathmc` builds the chain as a structural causal model and computes **path-specific effects** by propagating posterior draws through the intervened graph — the modern, uncertainty-aware mediation analysis. 7-step contract.

1.1 2 · Simulate a ground truth

We simulate the funnel as a **causal chain**: `onboarding_score -> engagement -> activated -> converted`, with `channel_quality` feeding both engagement and activation (a common driver we'll need to account for). Each arrow is a real structural equation with a planted coefficient, so we know the **true** total effect of onboarding on conversion *and how it splits across paths* — which is exactly what we'll ask the model to recover. By construction most of onboarding's effect flows the long way round (`engagement -> activated -> converted`), with a smaller slice going `engagement -> converted` directly. If the method can rediscover that split from data alone, we can trust it on a funnel where we *don't* know the answer.

```
TRUE effects of onboarding on conversion:
  indirect_via_engagement_activated    3.17
  indirect_via_engagement_only         0.77
  total                                3.94
```

	onboarding_score	channel_quality	engagement	activated	converted
0	0.864798	0.726815	2.576471	3.869429	4.870321
1	0.855303	0.790067	3.206651	5.673260	6.371102
2	0.811023	0.348493	2.493785	4.691864	5.394113
3	0.261446	0.465793	0.572682	1.261547	1.264003
4	0.077199	0.887974	1.520775	3.259916	3.146544

1.2 3 · Identify — direct vs indirect effects (NDE / NIE)

For a chain $X \to M \to Y$ the **total effect** decomposes into a **natural direct effect** and **natural indirect effect** (Pearl):

$$\mathrm{TE} = \mathrm{NDE} + \mathrm{NIE}, \quad \mathrm{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \mathrm{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Under a linear SCM these are the **product-of-coefficients**: indirect $= a \cdot b$, direct $= c$, and `pathmc` computes each path-specific effect as exactly that product along the path (exact for the linear SCM here — not a Monte-Carlo simulation through the graph). The **proportion mediated** = NIE/TE tells you how much of onboarding's value is "really" downstream engagement/activation. Crucially we **include a direct onboarding->converted edge in the model and test it**, so a near-100%-mediated answer is *discovered* (the coefficient comes out ~ 0), not forced by leaving the edge out.

Assumptions (stronger than a single-link analysis): no unmeasured confounding of $X - Y$, $M - Y$, *and* $X - M$, plus the **cross-world** condition that no $M - Y$ confounder is itself affected by X. Natural effects are *not identified* if that last one fails — and the $M - Y$ confounding assumption is the one most likely to break, so we stress-test it in step 6.

1.3 4 · Estimate — fit the chain and query every path

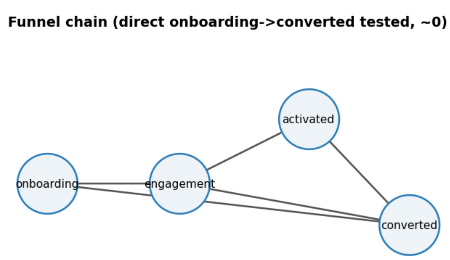
We hand `pathmc` the **whole chain at once** — one equation per stage — rather than three disconnected regressions. That matters: because the model knows the full graph, it can propagate an intervention on onboarding *through* engagement and activation to conversion, giving genuine path-specific effects instead of naive stage-by-stage correlations. The `effects_summary()` below lists every structural coefficient (the a's and b's from Step 3); Step 5 then combines them into total, direct, and indirect effects with uncertainty.

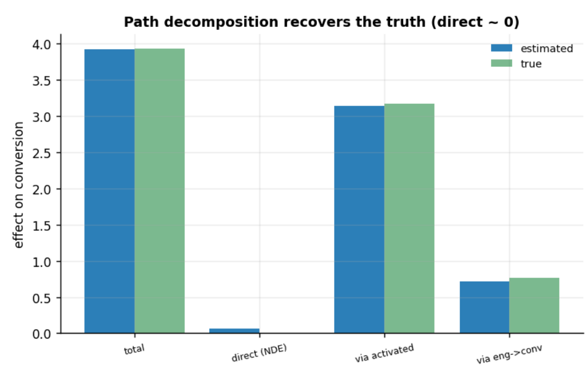
name	mean	sd	hdi_3%	hdi_97%
e_on	2.152	0.046	2.068	2.239
e_ch	1.130	0.047	1.051	1.225
a_eng	1.618	0.019	1.583	1.654
a_ch	0.589	0.052	0.492	0.687
c_act	0.902	0.021	0.861	0.941
c_eng	0.333	0.044	0.255	0.420
c_on	0.061	0.055	-0.044	0.155

1.4 5 · Validate — recover the total, the paths, the DIRECT effect, and the proportion mediated

We query four things and check each against the planted truth: the **total** effect of onboarding on conversion; the effect flowing through each **path**; the **direct** (natural direct) effect — which the model is now free to find, and which should come out $\sim$ **0** if onboarding really works only through the funnel; and the **proportion mediated** (indirect ÷ total). A proportion near 100% *with a direct effect whose interval includes 0* is the honest version of "onboarding works almost entirely by feeding the rest of the funnel" — a very different investment story from a direct effect.

```
TOTAL onboarding -> converted           3.92  (true 3.94)  [90% 3.76, 4.09]
  DIRECT (NDE) onboarding -> converted  0.06  (true 0.00)  [90% -0.03, 0.15]
  via engagement->activated->conv       3.14  (true 3.17)  [90% 2.97, 3.32]
  via engagement->converted             0.72  (true 0.77)  [90% 0.56, 0.87]
The DIRECT effect's 90% interval includes 0, so the ~98% proportion mediated is
DISCOVERED - the model was free to find a direct onboarding->conversion path and
the data say there isn't one.
```





How to read this. *Left* is the causal graph we fit — including a **direct** onboarding->conversion arrow, which we deliberately added so the model could find a direct effect if one existed. *Right* —

the estimated total and each path effect (blue) sit on top of the planted truths (green), and the **direct (NDE) bar is $\sim$ 0** (its interval includes zero): onboarding works almost entirely *through* the funnel, and that near-100%-mediated result is now **discovered, not assumed**. This is the decomposition a naive "regress conversion on each stage" analysis cannot give you.

1.5 6 · Decide, in euros — a *rankable* leverage metric, and its fragility

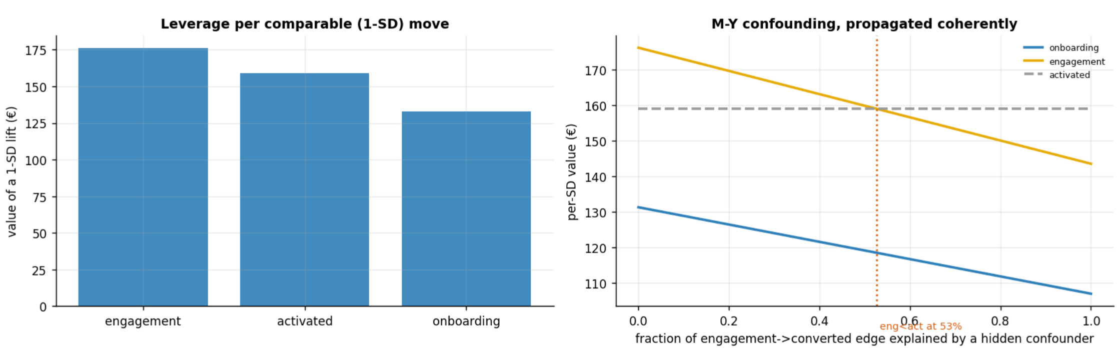
Two moves, and both correct a trap.

(1) Rank the levers — but not by "€ per one-unit change." A one-unit move means something completely different per stage (one unit of `onboarding_score` is a large fraction of its whole range; one unit of `activated` is a fraction of a standard deviation), so a €/unit ranking is not a rankable quantity — it *reverses* when you switch to comparable **per-standard-deviation** moves. The decision-relevant metric is **ROI = (€ value of a 1-SD lift) ÷ (€ cost to produce that 1-SD lift)**; we state an illustrative equal cost per SD and report `P(ROI > 1)`. Swap in your real per-stage costs and the ranking follows.

(2) Stress-test the split — coherently. The direct/indirect decomposition rests on *no unmeasured confounding of the mediator–outcome link* (Step 3). A confounder of engagement->conversion biases the fitted edge — but that same edge carries **both** the engagement lever **and** onboarding's total effect, so an honest sweep must shrink them *together* (a common mistake is to bias one and freeze the other). We do that per posterior draw and read off how much of the engagement->conversion edge a hidden confounder would have to explain before the leverage ranking changes.

lever	€/unit	€/SD	ROI (€/€)	P(ROI>1)
engagement	215	176	1.76	1.0
activated	108	159	1.59	1.0
onboarding	470	133	1.33	1.0

```
Per-UNIT vs per-SD DISAGREE: a 1-unit onboarding move is only 0.28 SD (tiny)
while a
1-unit activated move is 1.47 SD (large), so €/unit is not rankable - use ROI on
a stated
cost. At equal €100/SD the order is engagement > activated > onboarding and all
clear break-even.
Fragility: engagement's per-SD lead over activation vanishes once a hidden M-Y
confounder explains ~53% of the engagement->converted edge - a real caveat.
Onboarding (the root cause) stays robust.
```



1.6 7 · Caveats

- **Natural effects lean on strong assumptions** — no unmeasured confounding of any of the three links plus the cross-world condition. Step 6 quantified *coherently* (shrinking both affected levers together) how much $M - Y$ confounding would flip the leverage ranking; the point estimate alone hides that the engagement-vs-activation ordering is the fragile one.
- **A €/unit leverage ranking is not a rankable quantity** when the stage metrics live on different scales — it reverses under per-SD units. Decide on **ROI against a stated cost-to-move**, not €/unit.
- **Don't control away the mechanism.** The goal is to *decompose* the path; adjusting for a mediator when you wanted the total effect removes the very thing you're pricing (nb 05).
- **Linear-SCM decomposition is exact only if the SCM is linear.** For strong nonlinearities lean on `pathmc`'s numbers, not hand-multiplied coefficients.